

Full Stack Development with MERN

Project Documentation format

1. Introduction

- Project Title: [SHOP EZ e-commerce]

2. Project Overview: ShopEz is a full-stack e-commerce platform designed for students and small businesses that need an affordable and easy-to-deploy online store solution. The project demonstrates real-world full-stack development skills while providing essential e-commerce functionalities.

Goals

- Provide an affordable e-commerce solution.
- Enable small businesses to sell products online.
- Demonstrate modern web development practices.

2. Features

Key Features

- User Registration and Login
- Secure Authentication
- Product Browsing and Search
- Product Categories and Filtering

- Shopping Cart Management
 - Secure Checkout Process
 - Online Payment Integration
 - Order Tracking and History
 - Email Notifications
 - Admin Product Management
 - Inventory Management
 - Sales Reports and Analytics
 - Responsive Design for Desktop and Mobile Devices
-

3. Architecture

Frontend Architecture

The frontend is developed using React.js and follows a component-based architecture.

Components:

- Login & Registration Components
- Product Listing Component
- Product Details Component
- Shopping Cart Component
- Checkout Component
- Order History Component

- Admin Dashboard Component

Technologies:

- React.js
- HTML5
- CSS3
- JavaScript
- Bootstrap

Backend Architecture

The backend follows a RESTful API architecture using Node.js and Express.js.

Modules:

- Authentication Module
- Product Management Module
- Cart Management Module
- Order Processing Module
- Payment Processing Module
- Notification Module

Technologies:

- Node.js
- Express.js
- JWT Authentication

- Bcrypt Password Encryption

Database

MongoDB is used to store application data.

Collections:

- Users
- Products
- Orders
- Cart
- Payments
- Reviews

Database Operations:

- Create
- Read
- Update
- Delete (CRUD)

4. Setup Instructions

Prerequisites

- Node.js (v18 or later)
- MongoDB
- Git

- VS Code

Installation

Clone Repository

```
git clone https://github.com/username/shopez.git
```

```
cd shopez
```

Install Frontend Dependencies

```
cd client
```

```
npm install
```

Install Backend Dependencies

```
cd ../server
```

```
npm install
```

Environment Variables

Create a .env file:

```
PORT=5000
```

```
MONGODB_URI=your_mongodb_connection_string
```

```
JWT_SECRET=your_secret_key
```

```
RAZORPAY_KEY=your_key
```

5. Folder Structure

Client (Frontend)

client/

```
└── public/
└── src/
    ├── components/
    ├── pages/
    ├── services/
    ├── context/
    ├── App.js
    └── index.js
```

Server (Backend)

```
server/
└── controllers/
└── models/
└── routes/
└── middleware/
└── config/
└── server.js
└── package.json
```

6. Running the Application

Start Frontend

cd client

npm start

Frontend URL:

<http://localhost:3000>

Start Backend

cd server

npm start

Backend URL:

<http://localhost:5000>

7. API Documentation

Authentication APIs

Register User

POST /api/auth/register

Request:

```
{  
  "name": "Anshi",  
  "email": "user@gmail.com",  
  "password": "123456"  
}
```

Response:

```
{  
  "message": "User Registered Successfully"  
}
```

Login User

POST /api/auth/login

Response:

```
{  
  "token": "jwt_token"  
}
```

Product APIs

Get All Products

GET /api/products

Add Product

POST /api/products

Update Product

PUT /api/products/:id

Delete Product

DELETE /api/products/:id

Order APIs

POST /api/orders

GET /api/orders

GET /api/orders/:id

Cart APIs

POST /api/cart/add

DELETE /api/cart/remove

GET /api/cart

8. Authentication

ShopEz uses JWT-based authentication.

Security Features:

- JWT Token Authentication
- Password Hashing using Bcrypt
- Protected Routes
- Role-Based Access Control
- HTTPS Communication

Authentication Flow:

1. User logs in.
2. Server validates credentials.

3. JWT token is generated.
 4. Token is stored in browser storage.
 5. Protected APIs validate the token.
-

G. User Interface

The application provides:

- Home Page
 - Login Page
 - Registration Page
 - Product Catalog
 - Product Details Page
 - Shopping Cart
 - Checkout Page
 - Order History
 - Admin Dashboard
-

10. Testing

Testing Strategy

- Functional Testing
- Integration Testing
- User Acceptance Testing

- API Testing

Tools Used

- Postman
- Chrome Developer Tools
- Manual Testing

11. Screenshots / Demo

The image displays two screenshots of the ShopEZ web application interface.

Top Screenshot: Login Page

- Header:** ShopEZ logo, Browse Products link, shopping cart icon, Login link, and Sign Up button.
- Form:**
 - Welcome Back to ShopEZ:** Log in to manage your private cart, place orders, and browse dynamic catalogs.
 - EMAIL ADDRESS:** Input field containing you@example.com.
 - PASSWORD:** Input field with placeholder text Enter your security password.
 - Sign In:** A blue button with a right arrow icon.
- Footer:** Don't have an retail account yet? Register a free account. Testing Hint: Admin permissions can be tested with standard admin credentials, or log in with any user account.

Bottom Screenshot: Registration Page

- Header:** ShopEZ logo, Browse Products link, shopping cart icon, Login link, and Sign Up button.
- Form:**
 - Create an Account on ShopEZ:** Sign up for free to access a cart, log order histories, and execute checkouts.
 - FULL NAME:** Input field containing Mohan.
 - EMAIL ADDRESS:** Input field containing mohan@gmail.com.
 - PASSWORD:** Input field with masked characters (dots).
 - CONFIRM PASSWORD:** Input field with masked characters (dots).
 - Create Account:** A blue button with a right arrow icon.

ShopEZ

[Browse Products](#)

[My Orders](#)



Mohan



SUMMER SEASON OUTLET

Minimalist Design, Maximum Performance.

Discover our curated essentials designed for standard lifestyle perfection. Shop secure, get fast shipping.

Explore Catalog

Our Products

Showing 6 premium essentials

 Search catalog...

11 Default Sort

ShopEZ

[Browse Products](#)

[My Orders](#)



Mohan



ALL

ELECTRONICS

FOOTWEAR

CLOTHING

ELECTRONICS



iPhone 15

Latest Apple iPhone with A16 chip

₹79999.00

Stock: 44 left

Add

ELECTRONICS



Samsung Galaxy S24

Latest Samsung flagship with AI features

₹74999.00

Stock: 30 left

Add

ELECTRONICS



MacBook Air M2

Apple MacBook with M2 chip, 8GB RAM, 256GB SSD

₹114999.00

Stock: 20 left

Add

ELECTRONICS



Sony WH-1000XM5

Industry leading noise cancelling wireless headphones

₹29999.00

Stock: 45 left

Add

FOOTWEAR



Nike Air Max

Comfortable running shoes with air cushioning

₹8999.00

Stock: 78 left

Add

CLOTHING



Levi's 511 Jeans

Slim fit jeans in stretch denim

₹2999.00

Stock: 148 left

Add

ShopEZ

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VISA

MC

AMEX

STRIPE

ShopEZ

Browse Products

My Orders

1

Mohan

user

iPhone15

Lifestyle

QUANTITY

1

TOTAL PRICE

₹79999.00

←

Continue Shopping

Order Summary

%

Promo Code

Apply

Subtotal

₹79999.00

Shipping Estimate

FREE

Estimated Tax (8%)

₹6399.92

Grand Total

₹86398.92

Proceed to Checkout

→

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ShopEZ

Browse Products

My Orders

Mohan

user

Your Orders

Review, trace, and audit purchase logs and shipping operations.

#

ORDER REF

6a34399abdf16460d0d3c583

DATE LOGGED

Jun 19, 2026

₹

TOTAL CHARGED

₹79999.00

QUANTITY

1 item inside

Unpaid

Pending

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VISA

MC

AMEX

STRIPE

12. Known Issues

- Search performance may decrease with a very large product database.
 - Email delivery may be delayed due to SMTP restrictions.
 - Mobile responsiveness may require additional optimization on older devices.
 - Payment gateway testing mode may not support all transaction types.
-

13. Future Enhancements

- AI-based Product Recommendation System
- Wishlist Functionality
- Multi-Vendor Marketplace Support
- Mobile Application Development
- Advanced Analytics Dashboard
- Chatbot Customer Support
- Multiple Payment Gateway Integration
- Real-Time Order Tracking
- Product Reviews and Ratings
- Coupon and Discount Management System